Table 1: Fixed-method AUROC/AP by dataset, design and positive-label budget. Each entry averages 20 repeated splits using final refitted models; no dataset or method is selected according to performance.

Design	Budget	Method	AUROC	AP	n
Credit Default					
OS	0.05	Logistic probability correction	0.7025	0.4494	20
OS	0.05	nnPU logistic	0.7033	0.4588	20
OS	0.05	Naive CatBoost	0.7529	0.5046	20
OS	0.05	CS CatBoost	0.7461	0.4936	20
OS	0.05	RN CatBoost	0.7137	0.4566	20
OS	0.05	PU gradient boosting	0.6954	0.4209	20
OS	0.20	Logistic probability correction	0.7203	0.4878	20
OS	0.20	nnPU logistic	0.7182	0.4930	20
OS	0.20	Naive CatBoost	0.7695	0.5332	20
OS	0.20	CS CatBoost	0.7695	0.5343	20
OS	0.20	RN CatBoost	0.7242	0.4780	20
OS	0.20	PU gradient boosting	0.7399	0.5053	20
TS	0.05	Logistic probability correction	0.7007	0.4532	20
TS	0.05	nnPU logistic	0.7030	0.4667	20
TS	0.05	Naive CatBoost	0.7533	0.5102	20
TS	0.05	CS CatBoost	0.7483	0.5030	20
TS	0.05	RN CatBoost	0.7153	0.4582	20
TS	0.05	PU gradient boosting	0.7466	0.4927	20
TS	0.20	Logistic probability correction	0.7168	0.4854	20
TS	0.20	nnPU logistic	0.7154	0.4927	20
TS	0.20	Naive CatBoost	0.7694	0.5356	20
TS	0.20	CS CatBoost	0.7677	0.5320	20
TS	0.20	RN CatBoost	0.7178	0.4695	20
TS	0.20	PU gradient boosting	0.7591	0.5203	20
Dry Bean					
OS	0.05	Logistic probability correction	0.9853	0.9625	20
OS	0.05	nnPU logistic	0.9836	0.9595	20
OS	0.05	Naive CatBoost	0.9853	0.9515	20
OS	0.05	CS CatBoost	0.9837	0.9418	20
OS	0.05	RN CatBoost	0.9874	0.9647	20
OS	0.05	PU gradient boosting	0.9588	0.8938	20
OS	0.20	Logistic probability correction	0.9896	0.9720	20
OS	0.20	nnPU logistic	0.9890	0.9713	20
OS	0.20	Naive CatBoost	0.9890	0.9639	20
OS	0.20	CS CatBoost	0.9890	0.9642	20
OS	0.20	RN CatBoost	0.9900	0.9699	20
OS	0.20	PU gradient boosting	0.9849	0.9528	20
TS	0.05	Logistic probability correction	0.9847	0.9605	20
TS	0.05	nnPU logistic	0.9833	0.9588	20
TS	0.05	Naive CatBoost	0.9808	0.9300	20
TS	0.05	CS CatBoost	0.9763	0.9104	20
TS	0.05	RN CatBoost	0.9879	0.9638	20
TS	0.05	PU gradient boosting	0.9737	0.9253	20
TS	0.20	Logistic probability correction	0.9895	0.9718	20
TS	0.20	nnPU logistic	0.9887	0.9715	20
TS	0.20	Naive CatBoost	0.9889	0.9621	20
TS	0.20	CS CatBoost	0.9874	0.9561	20
TS	0.20	RN CatBoost	0.9892	0.9680	20
TS	0.20	PU gradient boosting	0.9864	0.9565	20

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Fixed-method real-data results (continued)

Design	Budget	Method	AUROC	AP	n
HTRU2					
OS	0.05	Logistic probability correction	0.9606	0.8774	20
OS	0.05	nnPU logistic	0.9661	0.8919	20
OS	0.05	Naive CatBoost	0.9633	0.8511	20
OS	0.05	CS CatBoost	0.9600	0.8334	20
OS	0.05	RN CatBoost	0.9656	0.8886	20
OS	0.05	PU gradient boosting	0.9238	0.7996	20
OS	0.20	Logistic probability correction	0.9627	0.8902	20
OS	0.20	nnPU logistic	0.9713	0.9060	20
OS	0.20	Naive CatBoost	0.9705	0.8893	20
OS	0.20	CS CatBoost	0.9712	0.8855	20
OS	0.20	RN CatBoost	0.9710	0.8928	20
OS	0.20	PU gradient boosting	0.9425	0.8476	20
TS	0.05	Logistic probability correction	0.9611	0.8773	20
TS	0.05	nnPU logistic	0.9644	0.8932	20
TS	0.05	Naive CatBoost	0.9633	0.8658	20
TS	0.05	CS CatBoost	0.9618	0.8559	20
TS	0.05	RN CatBoost	0.9638	0.8873	20
TS	0.05	PU gradient boosting	0.9329	0.8328	20
TS	0.20	Logistic probability correction	0.9663	0.8968	20
TS	0.20	nnPU logistic	0.9702	0.9058	20
TS	0.20	Naive CatBoost	0.9696	0.8826	20
TS	0.20	CS CatBoost	0.9712	0.8893	20
TS	0.20	RN CatBoost	0.9707	0.8979	20
TS	0.20	PU gradient boosting	0.9542	0.8536	20
Online Shoppers					
OS	0.05	Logistic probability correction	0.6454	0.2395	20
OS	0.05	nnPU logistic	0.6402	0.2337	20
OS	0.05	Naive CatBoost	0.6981	0.2741	20
OS	0.05	CS CatBoost	0.6878	0.2674	20
OS	0.05	RN CatBoost	0.6913	0.2770	20
OS	0.05	PU gradient boosting	0.6344	0.2207	20
OS	0.20	Logistic probability correction	0.7085	0.2881	20
OS	0.20	nnPU logistic	0.7066	0.2844	20
OS	0.20	Naive CatBoost	0.7363	0.3210	20
OS	0.20	CS CatBoost	0.7351	0.3198	20
OS	0.20	RN CatBoost	0.7268	0.3131	20
OS	0.20	PU gradient boosting	0.7007	0.2729	20
TS	0.05	Logistic probability correction	0.6558	0.2490	20
TS	0.05	nnPU logistic	0.6526	0.2449	20
TS	0.05	Naive CatBoost	0.7086	0.2787	20
TS	0.05	CS CatBoost	0.6973	0.2724	20
TS	0.05	RN CatBoost	0.6993	0.2837	20
TS	0.05	PU gradient boosting	0.6937	0.2627	20
TS	0.20	Logistic probability correction	0.7109	0.2858	20
TS	0.20	nnPU logistic	0.7076	0.2813	20
TS	0.20	Naive CatBoost	0.7376	0.3161	20
TS	0.20	CS CatBoost	0.7359	0.3149	20
TS	0.20	RN CatBoost	0.7291	0.3146	20
TS	0.20	PU gradient boosting	0.7245	0.3043	20

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Fixed-method real-data results (continued)

Design	Budget	Method	AUROC	AP	n
Optical Digits					
OS	0.05	Logistic probability correction	0.8851	0.8742	20
OS	0.05	nnPU logistic	0.9020	0.8833	20
OS	0.05	Naive CatBoost	0.9389	0.9387	20
OS	0.05	CS CatBoost	0.9305	0.9318	20
OS	0.05	RN CatBoost	0.9338	0.9313	20
OS	0.05	PU gradient boosting	0.8415	0.8084	20
OS	0.20	Logistic probability correction	0.9351	0.9266	20
OS	0.20	nnPU logistic	0.9375	0.9215	20
OS	0.20	Naive CatBoost	0.9814	0.9816	20
OS	0.20	CS CatBoost	0.9789	0.9790	20
OS	0.20	RN CatBoost	0.9707	0.9703	20
OS	0.20	PU gradient boosting	0.9397	0.9409	20
TS	0.05	Logistic probability correction	0.8910	0.8814	20
TS	0.05	nnPU logistic	0.9016	0.8804	20
TS	0.05	Naive CatBoost	0.9382	0.9383	20
TS	0.05	CS CatBoost	0.9351	0.9364	20
TS	0.05	RN CatBoost	0.9373	0.9344	20
TS	0.05	PU gradient boosting	0.8518	0.8135	20
TS	0.20	Logistic probability correction	0.9301	0.9213	20
TS	0.20	nnPU logistic	0.9327	0.9085	20
TS	0.20	Naive CatBoost	0.9782	0.9784	20
TS	0.20	CS CatBoost	0.9763	0.9770	20
TS	0.20	RN CatBoost	0.9679	0.9667	20
TS	0.20	PU gradient boosting	0.9181	0.9092	20
Phishing Websites					
OS	0.05	Logistic probability correction	0.9292	0.9327	20
OS	0.05	nnPU logistic	0.9372	0.9404	20
OS	0.05	Naive CatBoost	0.9412	0.9382	20
OS	0.05	CS CatBoost	0.9417	0.9392	20
OS	0.05	RN CatBoost	0.9617	0.9649	20
OS	0.05	PU gradient boosting	0.9333	0.9284	20
OS	0.20	Logistic probability correction	0.9687	0.9703	20
OS	0.20	nnPU logistic	0.9642	0.9661	20
OS	0.20	Naive CatBoost	0.9762	0.9758	20
OS	0.20	CS CatBoost	0.9746	0.9740	20
OS	0.20	RN CatBoost	0.9809	0.9818	20
OS	0.20	PU gradient boosting	0.9661	0.9677	20
TS	0.05	Logistic probability correction	0.9274	0.9319	20
TS	0.05	nnPU logistic	0.9264	0.9304	20
TS	0.05	Naive CatBoost	0.9416	0.9386	20
TS	0.05	CS CatBoost	0.9436	0.9419	20
TS	0.05	RN CatBoost	0.9589	0.9624	20
TS	0.05	PU gradient boosting	0.9459	0.9403	20
TS	0.20	Logistic probability correction	0.9671	0.9699	20
TS	0.20	nnPU logistic	0.9587	0.9616	20
TS	0.20	Naive CatBoost	0.9718	0.9706	20
TS	0.20	CS CatBoost	0.9720	0.9707	20
TS	0.20	RN CatBoost	0.9806	0.9820	20
TS	0.20	PU gradient boosting	0.9631	0.9621	20

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Fixed-method real-data results (continued)

Design	Budget	Method	AUROC	AP	n
Rice					
OS	0.05	Logistic probability correction	0.9722	0.9645	20
OS	0.05	nnPU logistic	0.9682	0.9610	20
OS	0.05	Naive CatBoost	0.9530	0.9225	20
OS	0.05	CS CatBoost	0.9453	0.9084	20
OS	0.05	RN CatBoost	0.9693	0.9602	20
OS	0.05	PU gradient boosting	0.9369	0.9007	20
OS	0.20	Logistic probability correction	0.9759	0.9699	20
OS	0.20	nnPU logistic	0.9767	0.9711	20
OS	0.20	Naive CatBoost	0.9673	0.9487	20
OS	0.20	CS CatBoost	0.9621	0.9392	20
OS	0.20	RN CatBoost	0.9687	0.9549	20
OS	0.20	PU gradient boosting	0.9620	0.9396	20
TS	0.05	Logistic probability correction	0.9685	0.9603	20
TS	0.05	nnPU logistic	0.9706	0.9634	20
TS	0.05	Naive CatBoost	0.9519	0.9249	20
TS	0.05	CS CatBoost	0.9483	0.9211	20
TS	0.05	RN CatBoost	0.9667	0.9579	20
TS	0.05	PU gradient boosting	0.9446	0.9185	20
TS	0.20	Logistic probability correction	0.9733	0.9658	20
TS	0.20	nnPU logistic	0.9756	0.9698	20
TS	0.20	Naive CatBoost	0.9658	0.9473	20
TS	0.20	CS CatBoost	0.9639	0.9441	20
TS	0.20	RN CatBoost	0.9697	0.9575	20
TS	0.20	PU gradient boosting	0.9635	0.9398	20
Spambase					
OS	0.05	Logistic probability correction	0.8753	0.7979	20
OS	0.05	nnPU logistic	0.9122	0.8650	20
OS	0.05	Naive CatBoost	0.9260	0.8924	20
OS	0.05	CS CatBoost	0.9075	0.8686	20
OS	0.05	RN CatBoost	0.9473	0.9264	20
OS	0.05	PU gradient boosting	0.7732	0.6995	20
OS	0.20	Logistic probability correction	0.9323	0.8779	20
OS	0.20	nnPU logistic	0.9391	0.9076	20
OS	0.20	Naive CatBoost	0.9663	0.9499	20
OS	0.20	CS CatBoost	0.9634	0.9446	20
OS	0.20	RN CatBoost	0.9732	0.9612	20
OS	0.20	PU gradient boosting	0.9380	0.9155	20
TS	0.05	Logistic probability correction	0.8728	0.7990	20
TS	0.05	nnPU logistic	0.9041	0.8550	20
TS	0.05	Naive CatBoost	0.9310	0.9053	20
TS	0.05	CS CatBoost	0.9223	0.8940	20
TS	0.05	RN CatBoost	0.9490	0.9316	20
TS	0.05	PU gradient boosting	0.8485	0.7853	20
TS	0.20	Logistic probability correction	0.9277	0.8753	20
TS	0.20	nnPU logistic	0.9347	0.8982	20
TS	0.20	Naive CatBoost	0.9636	0.9443	20
TS	0.20	CS CatBoost	0.9583	0.9366	20
TS	0.20	RN CatBoost	0.9726	0.9603	20
TS	0.20	PU gradient boosting	0.9274	0.9028	20